Debarshi Chakraborty

LinkedIn GitHub Portfolio

Email: debarshi2933@gmail.com
Mobile: +91-903-8564-470

Experience

- **Associate Consultant** — July 2026 - Present
  Ernst & Young LLP — *Kolkata, West Bengal*
  - Delivered an **agentic supply planner** for a global CPG client's Digital Supply Planner program (**40+ countries**), eliminating **6,500+ hours** of manual planning effort.
  - Designed a **multi-agent architecture** orchestrating **10 sub-agents**, and authored and evaluated the underlying **Databricks Genie** agents.
  - Implemented the quantitative logic in **PySpark**, using **Bayesian MCMC** and simulation to reduce deadstock and run what-if inventory scenarios.
  - Added **long-term personalization** so agents remember user preferences, and **PPT generation** from any single response or an entire conversation.
  - Productionized the solution end to end with **Databricks Asset Bundles** for versioned, repeatable deployments across environments, cutting manual agent setup time by **80–90%**, and exposed **API endpoints** for application integration.
  - Developed **demand forecasting** models on time-series data using **Random Forest** and **XGBoost** with a **post-processing** layer, covering the **MLOps lifecycle** from backtesting and parallel runs against live forecasts to industrialized deployment on **Azure** and **Databricks**.
- **AI/ML Intern** — January 2026 - July 2026
  Rezolve Ai — *Kolkata, West Bengal*
  - Engineered a production **Agent-to-Agent (A2A)** multi-agent system end-to-end, including orchestrator design, **MCP** tool server deployment, and **Redis**-backed context sharing, powering a live retail conversational AI platform.
  - Developed a **LangMem**-backed long-term memory agent injecting past user preferences into the A2A flow, reducing redundant inputs by **25%**.
  - Refactored the prompt stack into a **3-layer architecture** managed in **Langfuse**, cutting prompt-related incidents by **30%** across **6+ sub-agents**.
  - Led evaluation of multiple **LLMs** for tool calling; through careful model selection, retained routing performance while lowering latency by over **40%**.
- **Summer Research Intern** — May 2025 - July 2025
  Indian Institute of Technology, Guwahati — *Guwahati, India*
  - Collaborated with **PhD researchers** on **graph algorithms** for the **Popular Matching Problem** in subcubic graphs and bounded-degree graph families.
  - Created and analyzed **combinatorial** and **reduction-based algorithms** to determine the existence of popular matchings under structural constraints; implemented prototypes in **Python**.

Projects

- **Mnemos: Agentic Memory System with Provenance-Tracked Consolidation**:
  - Created a self-hosted agentic memory system with a **LangGraph-orchestrated** "sleep cycle" for extracting, deduplicating, and contradiction-checking facts from episodic history.
  - Implemented **decay-ranked retrieval**, LLM-based contradiction resolution, and **EMA confidence updates**, preserving permanent provenance chains.
  - Shipped memory operations as **MCP tools** for Claude Desktop/Cursor with a React frontend for chat, memory inspection, and roadmap planning, and tested them against 20 multi-session eval scripts.
- **FinSight: Regulatory-Grade Multi-Agent Financial Intelligence System**:
  - Architected a **six-stage multi-agent pipeline** (Planner, Retriever, Analyst, Auditor, Comparator, Synthesizer) as a **LangGraph StateGraph** with hybrid BM25 + dense retrieval and cross-encoder reranking.
  - Engineered an entailment-checking **Auditor agent** that hard-blocks unverifiable claims and a deterministic currency normalizer that eliminates LLM arithmetic hallucination.
  - Added a **Qdrant-backed session and cross-session memory** layer that biases query planning while staying fully outside the claims/citations path, preserving full auditability.

- **Real-Time (T+0) Trade Settlement System for the US Market** :
  - Architected a 4-node permissioned blockchain network (**Hyperledger Besu**, **IBFT-2**) for atomic delivery-versus-payment settlement of tokenized securities.
  - Integrated **LangGraph** compliance/liquidity agents with SHAP/LIME explainability and **XGBoost** fraud detection, cutting counterparty risk by **91.7%**.
  - Reduced the settlement cycle from the industry-standard T+2 to real-time (T+0) via deterministic smart-contract execution, eliminating counterparty exposure windows.
- **Graph-Structured Deep Learning & Multi-Agent Decision Pipeline for Financial Forecasting** :
  - Proposed **THGNN-MaGNet**, a hybrid GNN (GRU-MAGE encoder, Sparse MoE, temporal-causal + global probabilistic hypergraphs) trained with a differentiable Spearman IC loss on **NIFTY 500**.
  - Achieved **rank-IC 0.0398** (+95% over baseline), **ICIR 0.420**, and **Sharpe 3.71** on a 583-day holdout backtest.
  - Orchestrated a **six-agent LangGraph pipeline** (GNN, sentiment, risk, macro, LLM report agents) with sub-45s end-to-end inference.

Education

- **Ramakrishna Mission Vivekananda Educational and Research Institute**
  Master of Science - Big Data Analytics *July 2024 - July 2026*
- **Ramakrishna Mission Vivekananda Centenary College**
  Bachelor of Science - Mathematics Honors *September 2021 - July 2024*

Skills Summary

- **Generative AI**: LLMs, RAG, Agentic AI, Multi-Agent Systems, MCP, Prompt Engineering, LLM Evaluation, LangGraph, LangChain, LangMem
- **ML & Statistics**: PyTorch, Scikit-learn, XGBoost, Random Forest, Time-Series Forecasting, Bayesian MCMC, GNNs
- **Data & Cloud**: Databricks, PySpark, MLflow, Azure, Azure OpenAI, GCP, Pandas, NumPy, Polars, Dask
- **MLOps & Infra**: Docker, FastAPI, Langfuse, Redis, Qdrant, Weaviate, Neo4j, Streamlit
- **Languages**: Python, SQL, R, Java, C++
- **Tools**: Git, GitHub, VS Code, Jupyter

Achievements

- **IEEE WCONF 2025 Publication** *2025*
  Published *"Real Time Trade Settlement for the US Market"* at the 3rd World Conference on Communication & Computing. DOI: 10.1109/WCONF64849.2025.11233239
- **Cleared IIT JAM 2024** *February 2024*
  Successfully cleared the IIT JAM 2024 examination in Mathematics, securing eligibility for master's programs at IITs.

Volunteer Experience

- **Event Lead: CodeCraft — Perceptron 26, RKMVERI Annual Tech Fest** *January 2026*
  Spearheaded the flagship Hackathon "CodeCraft", managing logistics, problem statements, and participant coordination for 300+ attendees.